

Experiment 4 — Join, subquery, and EXPLAIN results

Operation	Outcome
INNER JOIN	30 employee–department rows are returned.
LEFT JOIN	All 5 departments are returned, including a department that would have zero employees. In this data set, every department has employees.
Self-join	30 employee–manager rows are returned; Asha Mehta has no manager.
Three-way join	30 employee–department–project rows are returned.

Employees earning above their own department average

Department	Employee	Salary
Engineering	Asha Mehta	150,000.00
Engineering	Rohan Shah	110,000.00
Human Resources	Meera Iyer	95,000.00
Human Resources	Ishita Sen	72,000.00
Finance	Kabir Khan	105,000.00
Finance	Aditya Kulkarni	80,000.00
Sales	Nisha Gupta	100,000.00
Sales	Neel Kapoor	82,000.00
Operations	Leena Menon	88,000.00
Operations	Yash Chopra	71,000.00

EXISTS output

EXISTS output

Departments with at least one employee assigned to a project budgeted at 300,000 or more:

Department ID	Department
1	Engineering
4	Sales

Simulated set-operation output

Simulation	Result
INTERSECT	Engineering employees assigned to Engineering-owned projects: Asha Mehta, Rohan Shah, Arjun Rao, Priya Nair, Vikram Singh, Sneha Patel, and Dev Malhotra.
EXCEPT	Sales employees not assigned to a Sales-owned project: Harsh Sethi.

EXPLAIN comparison

Query shape	Expected plan outcome
department_id = 1	Can use idx_employees_department to look up matching employees.
department_id + 0 = 1	The expression prevents a direct indexed lookup, so the plan is less efficient.
Three-way join	Uses the employee department filter followed by primary-key lookups into Department and Project.
Correlated subquery	Evaluates the peer salary average per employee/department; it is more expensive than the direct indexed lookup.

Exact EXPLAIN cost estimates depend on the MySQL version, statistics, and server configuration.